Issue number: 5			
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Version	Date	Comments	Authors
1	15-10-2025	Start of issue	Michaela Edwinson
2	10-12-2025	Contact with RBR	Michaela Edwinson

1 Introduction

In Summer 2025, the TRIDENTE bio-optical sensor, manufactured by RBR, was installed on all VOTO *SAMBA ocean observatory* gliders to measure chlorophyll, backscatter, and phycocyanin. While at sea, a lag was discovered between up- and downcast in the phycocyanin data as per fig. 1.
This lag between up and downcast are seemingly connected to ambient temperature and appears in the majority missions with a larger effect where there is a warmer surface layer, labeled with issue description B in table 1). Some missions also display an abnormally high phycocyanin concentrations throughout the water column apart from this lag, labeled with issue description C in table 1). Sensors with serials 238733 (Issue description C, table 2) and 238740 (Issue description B, table 2) went through an in house fridge/freezer test which showed an impact on the data where recordings in lower temperatures resulted in an increase in the phycocyanin data, which is confirmed by RBR to be a calibration error. Read more about these tests in section 3. Unit 23832 is displaying a decrease of the highest values over time with decreasing temperature in October to November at sea (fig. 9, left) which is contradictory to the results from the workshop tests that showed the opposite behavior. Some sensors will be sent back to RBR and units that display the biggest error, with issue description C in table 1, is prioritized. Investigation is ongoing as we see temperature effects also on 23840 on the bench that recorded data of reasonable values at sea and as we gather understanding of the behaviors of the different units at sea and in the workshop. Evaluation will be made whether the data with issue description B (table 1) can be corrected for the thermal lag in post processing. Table 1 lists all missions with the Tridente with the phycocyanin channel mounted. The "Issue description" is explained in table 2 together with a quality control flag: suspect (3), failure (4) and good (1).

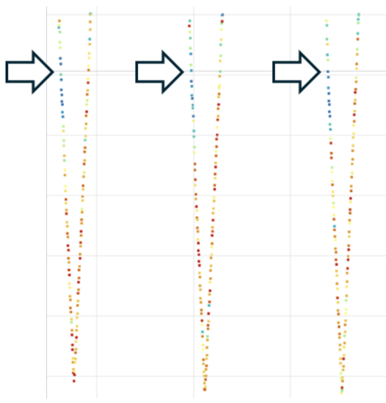

Figure 1: A lag between up- and downcast phycocyanin data was discovered at sea in August 2025.

Table 1: Details of missions affected by described issue. The different basins mentioned in the ‘Location’ column, follow the division made by HELCOM which is shown in fig.22.

Glider	Mission	Location	Mission Start Date	Tridente serial	Issue description
SEA045	100	Western Gotland Basin	2025-07-23	238736	**B**
	102	Åland Sea	2025-10-29	238742	**B**
SEA055	96	Bornholm Basin	2025-07-16	238734	**B**
	97	Bornholm Basin	2025-09-02	238734	**B**
	98	Bornholm Basin	2025-10-15	238734	**B**
SEA063	90	Eastern Gotland Basin	2025-07-24	238733	**C**
SEA066	67	Eastern Gotland Basin	2025-07-24	238739	**C**
SEA067	78	Bornholm Basin	2025-08-08	238738	**B**
	79	Bornholm Basin	2025-09-24	238738	**B**
	81	Bornholm Basin	2025-11-06	238738	**B**
SEA068	46	Skagerrak	2025-09-10	238740	**B**
	47	Skagerrak	2025-10-14	238740	**B**
SEA069	52	Skagerrak	2025-08-13	238741	**B**
	53	Skagerrak	2025-09-25	238741	**B**
	54	Skagerrak	2025-11-06	238741	**B**
SEA076	41	Åland Sea	2025-08-14	238743	**B**
	42	Eastern Gotland Basin	2025-11-11	238743	**B**
SEA077	44	Eastern Gotland Basin	2025-06-17	238732	**B**
	46	Eastern Gotland Basin	2025-08-28	238732	**B**
	47	Eastern Gotland Basin	2025-10-03	238732	**B**
SEA078	41	Eastern Gotland Basin	2025-08-27	238733	**C**
SEA079	36	Åland Sea	2025-09-25	238739	**C**

Table 2

Legend	Issue description	Flag
A	Data is good	1
B	Phycocyanin concentrations are displaying a lag between up and downcast. General unit behaviour and performance is under investigation and data is considered suspicious until proven otherwise.	3
C	Phycocyanin concentrations are displaying lag and abnormally high values. Due to a calibration error at manufacturer RBR.	4

2 Examples

2.1 Tridente data by location

Generally, chlorophyll and backscatter seem quite consistent between different units and phycocyanin data varies significantly between units (fig. 2, 3, 4, 5 and 8).

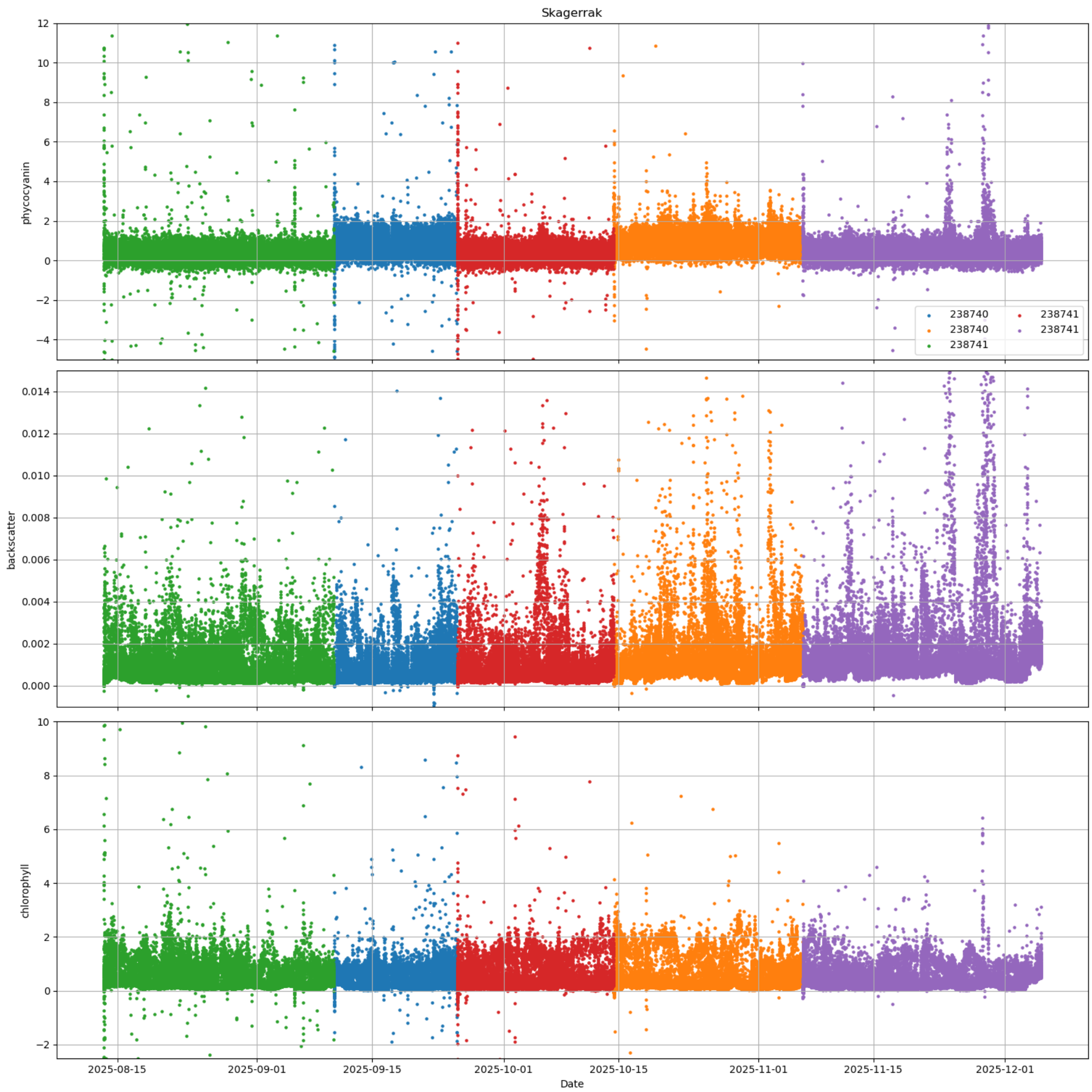

Figure 2: All missions with phycocyanin, backscatter and chlorophyll data recorded by the TRIDENTE in Skagerrak. Data is colored by sensor serial.

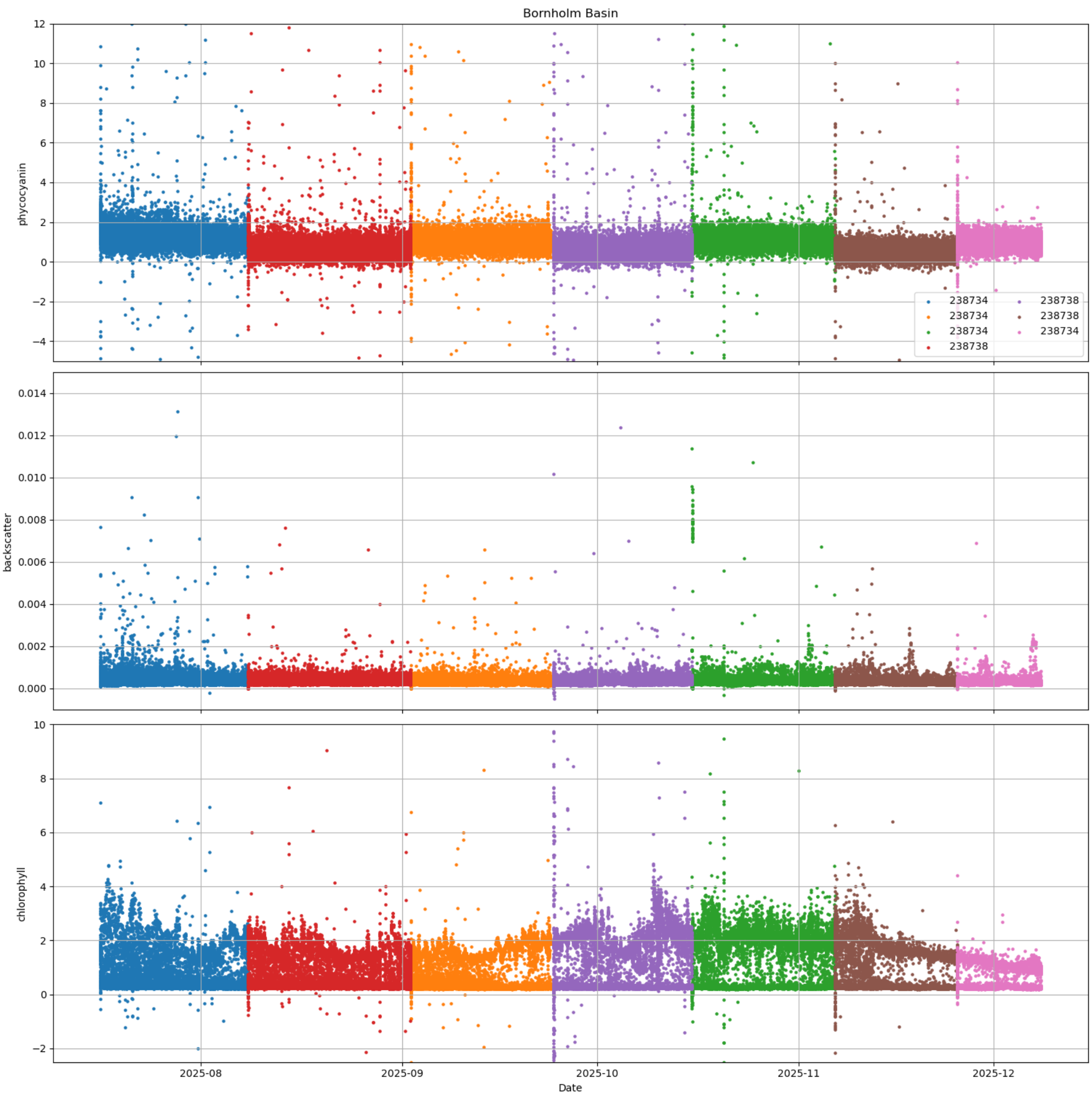

Figure 3: All missions with phycocyanin, backscatter and chlorophyll data recorded by the TRIDENTE in Bornholm Basin. Data is colored by sensor serial.

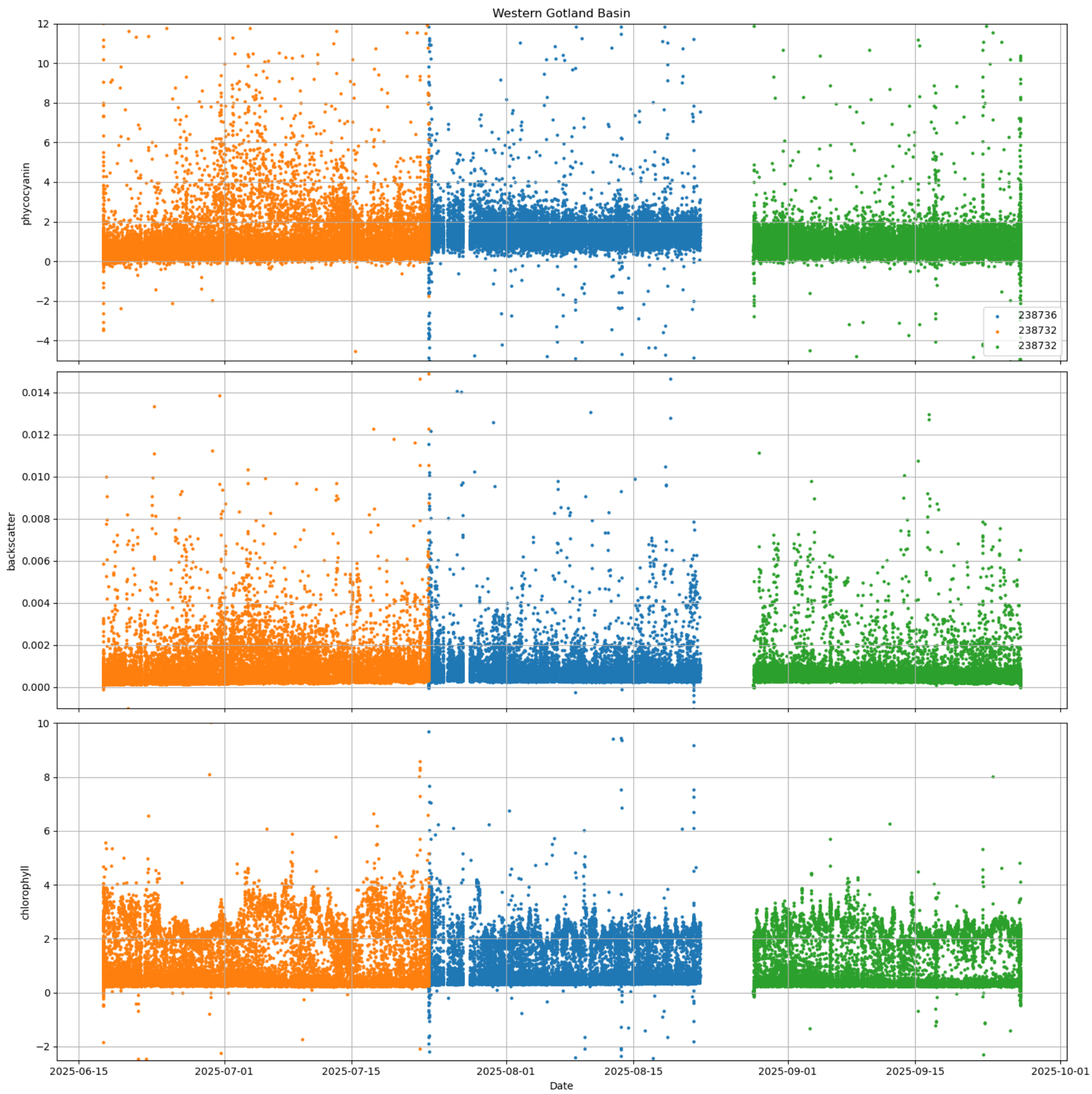

Figure 4: All missions with phycocyanin, backscatter and chlorophyll data recorded by the TRIDENTE in Western Gotland Basin. Data is colored by sensor serial.

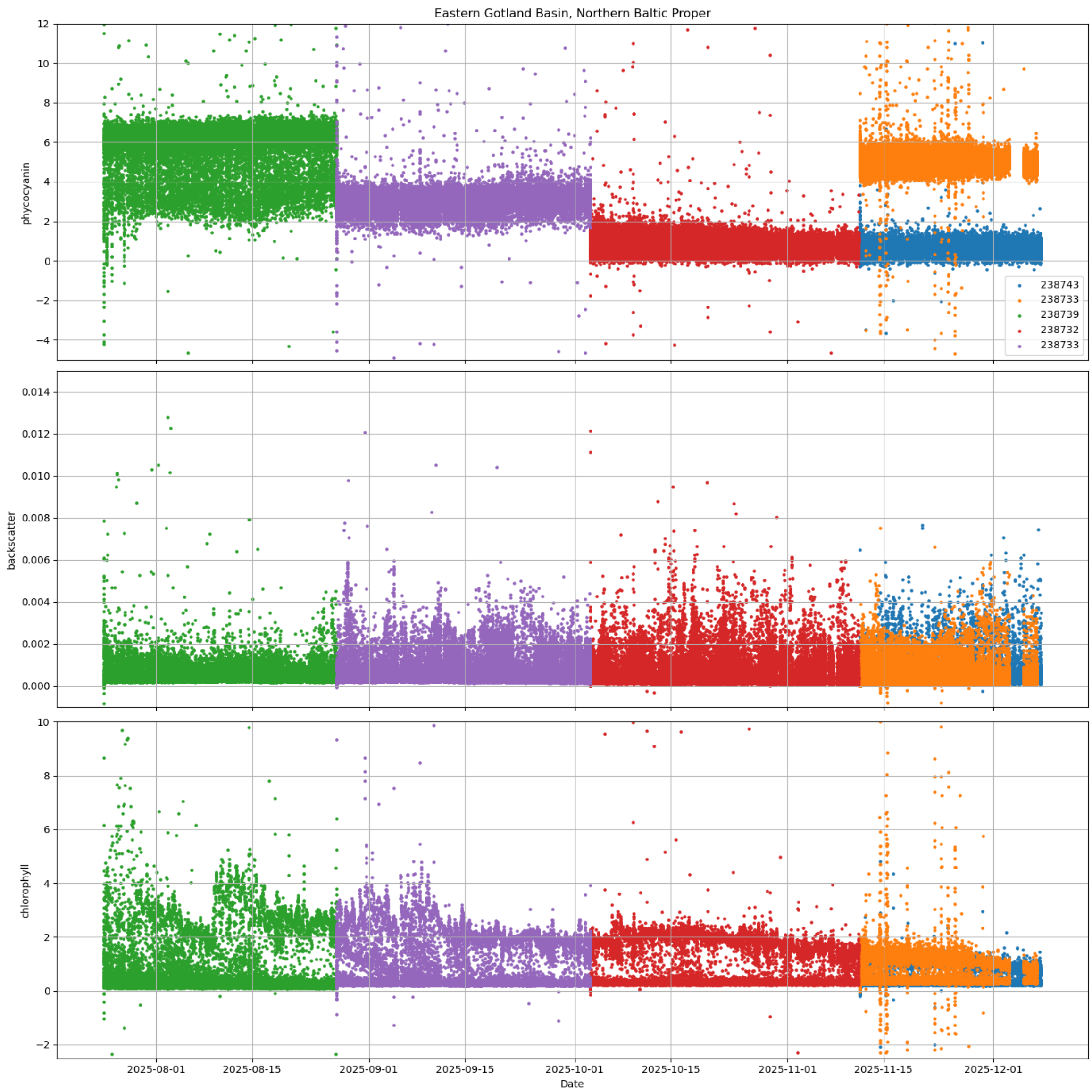

Figure 5: All missions with phycocyanin, backscatter and chlorophyll data recorded by the TRIDENTE in Eastern Gotland Basin. Data is colored by sensor serial.

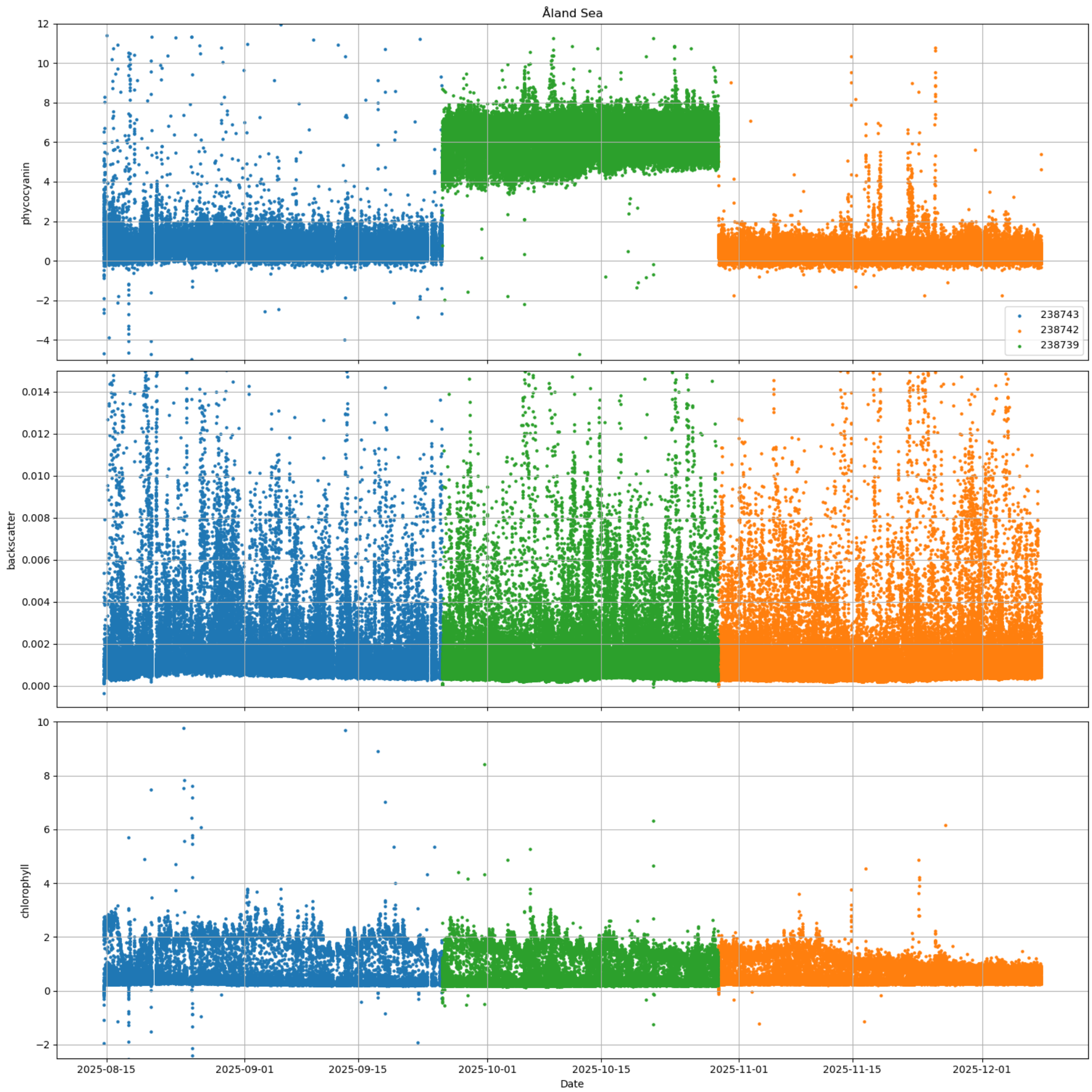

Figure 6: All missions with phycocyanin, backscatter and chlorophyll data recorded by the TRIDENTE in Åland Sea. Data is colored by sensor serial.

2.2 Thermal lag examples

Following examples displays the patterns of the Tridente phycocyanin data between up- and downcast of the units 238733 and 238740 which also went through the workshop tests as per section 3.

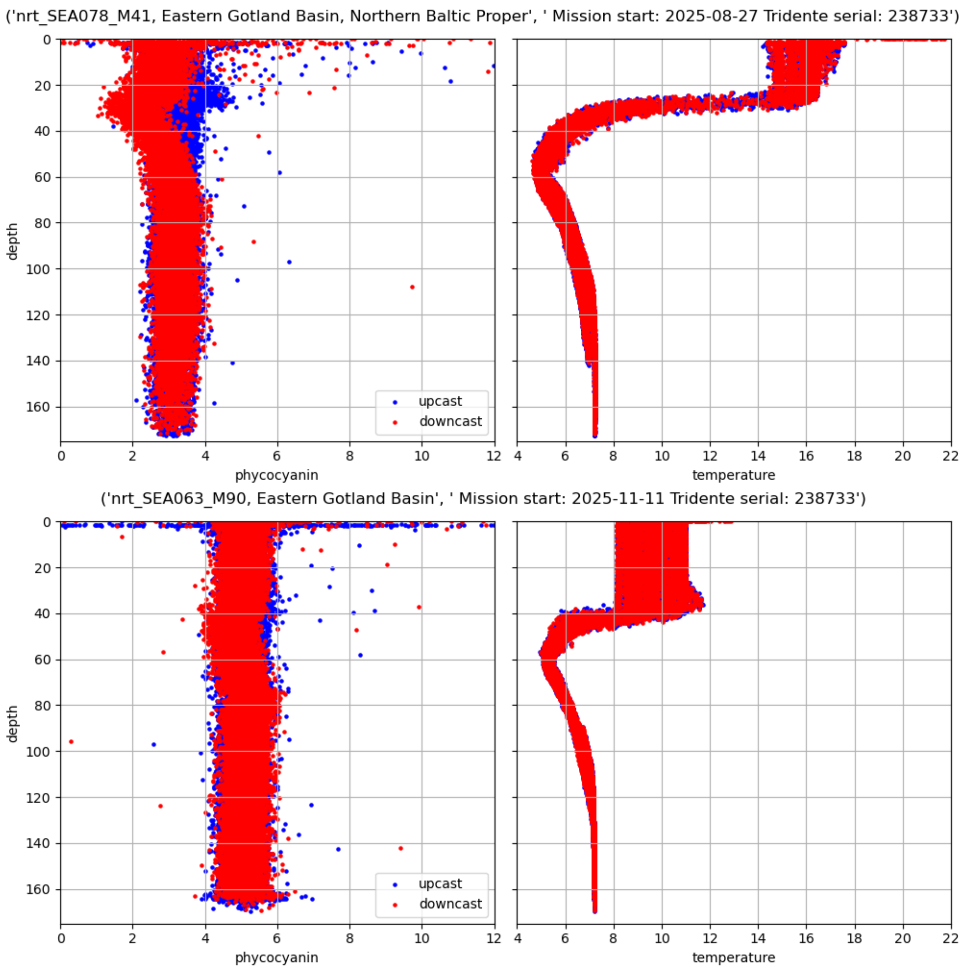

Figure 7: Phycocyanin vetical profiles from Tridente unit 238733 (left) and temperature (right) from Legato CTD colored by upcast (blue) and downcast (red) from a mission starting in August (upper row) and in November (lower row). Lag between up- and downcast data around 40 m depth is clearly increasing as there is a sharper temperature gradient as per the August data. The effect of temperature is confirmed in the workshop of this unit (fig. 20)

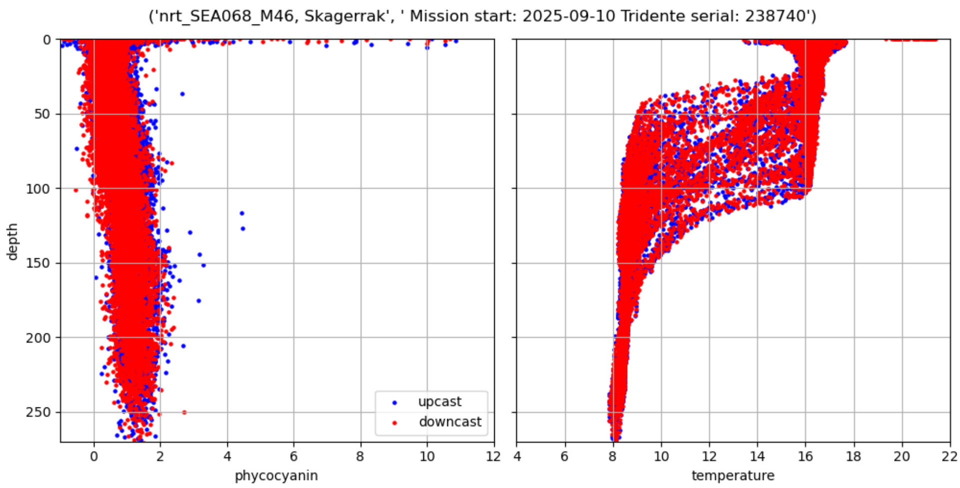

Figure 8: Phycocyanin vetical profiles (left) from Tridente unit 238740 and temperature (right) from Legato CTD colored by upcast (blue) and downcast (red) from a mission starting in September. Lag between up- and downcast data is not as clear at sea for unit 238740 although thermal effect is confirmed in the workshop (fig. 21), potentially due to lower true concentrations and a more dynamic temperature regime in the region.

2.3 Recorded Phycocyanin data of each Tridente unit

In the following examples, descriptions of any suspicious behaviors of each sensor are described in the corresponding figure text. If temperature affects all units in the same way is still to be investigated, however there is a possibility that units respond differently than according to the work shop tests, for example 238732 which upper values are decreasing with temperature October-November 2025 (fig. 9).

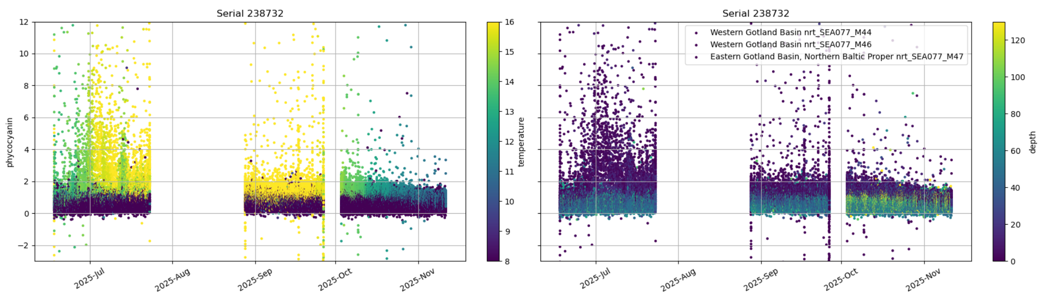

Figure 9: Phycocyanin recorded by Tridente 238732. Left plot is colored by temperature and right by depth. No expected blooms in the region in October according to *SMHI Algae monitoring data* and the October-November phycocyanin data decreasing with temperature is suspicious, behaving opposite to other investigated units. Data from this sensor was generally considered acceptable while at sea with only small lags between up- and downcast.

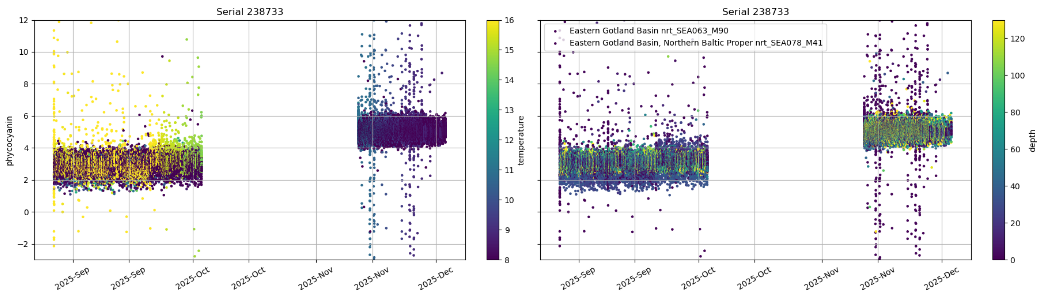

Figure 10: Phycocyanin recorded by Tridente 238733. Left plot is colored by temperature and right by depth. Different temperature in the different missions, with a significant difference between September and November missions where the November mission is showing significantly higher values in the colder waters. Thermal effect confirmed in the workshop (fig. 20).

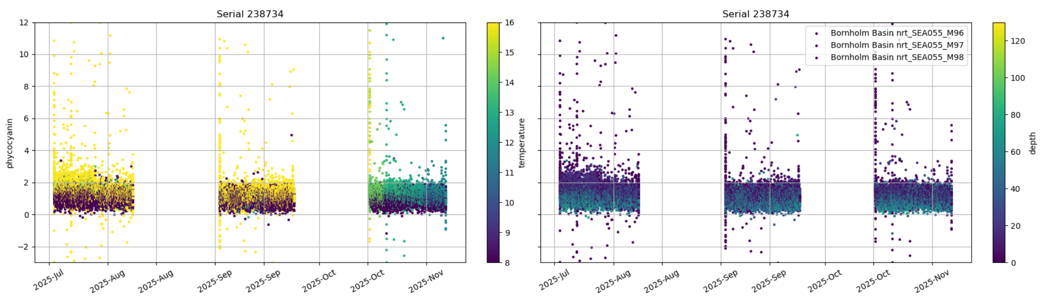

Figure 11: Phycocyanin recorded by Tridente 238734. Left plot is colored by temperature and right by depth.

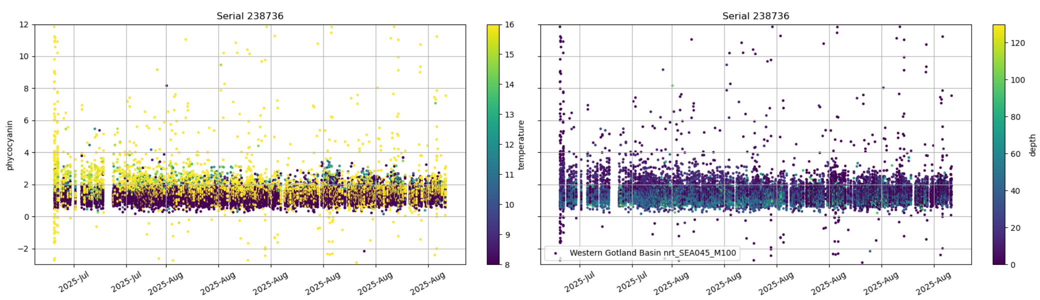

Figure 12: Phycocyanin recorded by Tridente 238736. Left plot is colored by temperature and right by depth.

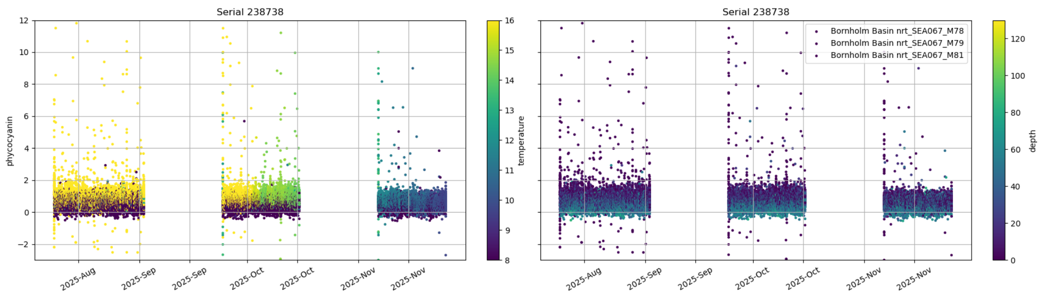

Figure 13: Phycocyanin recorded by Tridente 238738. Left plot is colored by temperature and right by depth. A decrease in the highest values of the surface waters in November, potentially due to decreasing temperature. No relevant algae bloom should be affecting these values according to *SMHI Algae monitoring data*.

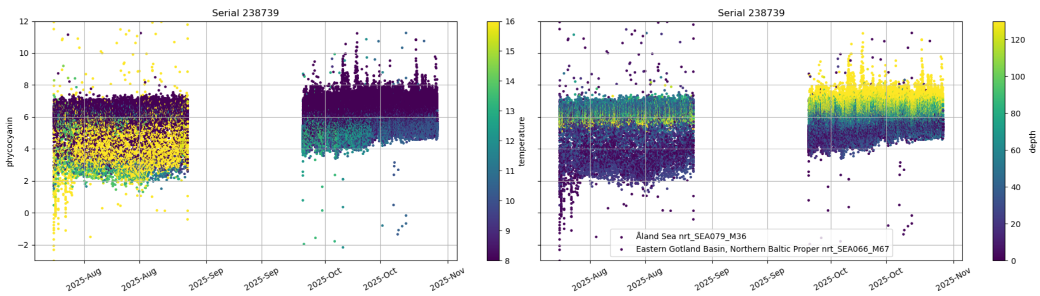

Figure 14: Phycocyanin recorded by Tridente 238739. Left plot is colored by temperature and right by depth. Abnormally high values of the mission overall with highest values recorded at depth. Lowest values of the October data seems to increase with decreasing temperatures. Due to a calibration error at RBR.

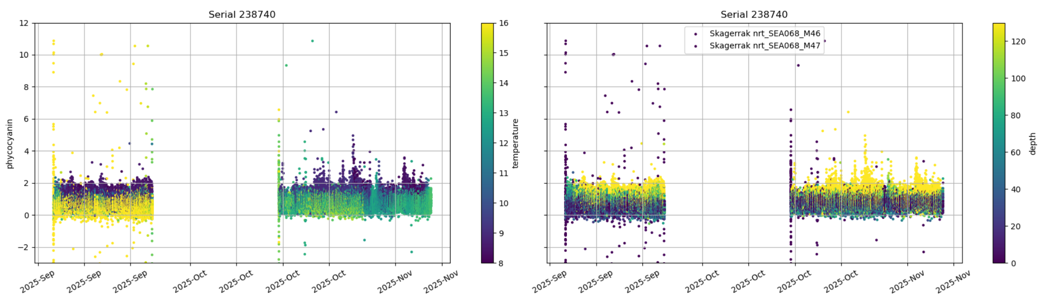

Figure 15: Phycocyanin recorded by Tridente 238740. Left plot is colored by temperature and right by depth. Thermal effect confirmed in the workshop (fig. 21) even if it is not evident at the missions at sea.

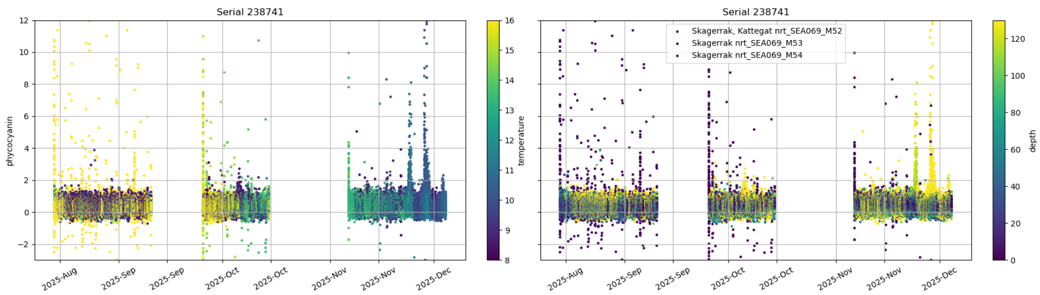

Figure 16: Phycocyanin recorded by Tridente 238741. Left plot is colored by temperature and right by depth.

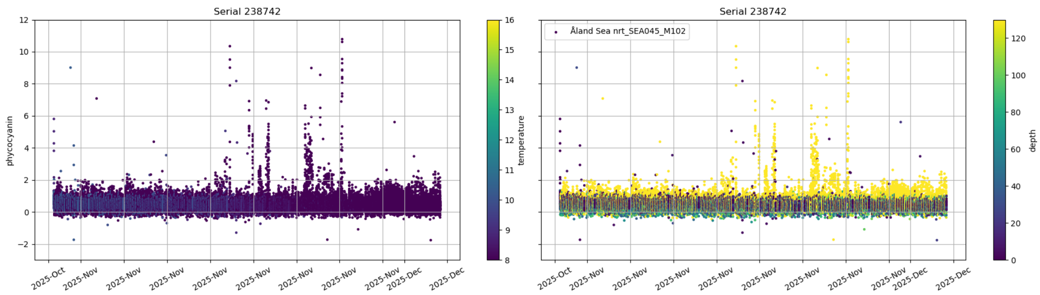

Figure 17: Phycocyanin recorded by Tridente 238742. Left plot is colored by temperature and right by depth.

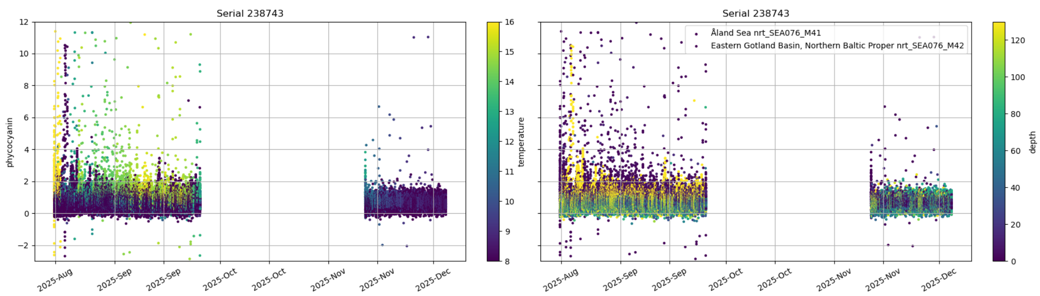

Figure 18: Phycocyanin recorded by Tridente 238743. Left plot is colored by temperature and right by depth.

3 TRIDENTE performance on bench in different ambient temperatures

After the lag and suspiciously high values were discovered at sea, an in house test with unit 23833 (Issue description C, table 2) and 23840 (Issue description B, table 2) was made. The units were placed to record data in different temperatures with the detectors covered as per figure 19. Temperatures varied between room temperature (20 °C), fridge (6 °C) and freezer (-18 °C). We see an increase in phycocyanin as ambient temperature gets lower while backscatter and chlorophyll does not show any suspicious response (figures 20 and 21). Upon returning Tridentes to room temperature, phycocyanin values exhibited a delayed recovery, generally requiring 30–60 minutes to return toward baseline values (at room temperature). The precise length of this lag, however, could not be conclusively defined within the scope of these tests.

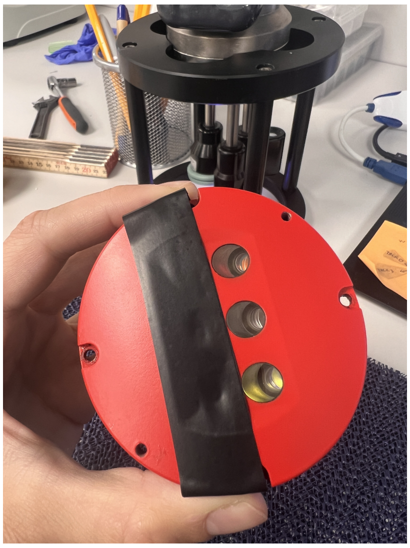

Figure 19: Tridente with covered detectors.

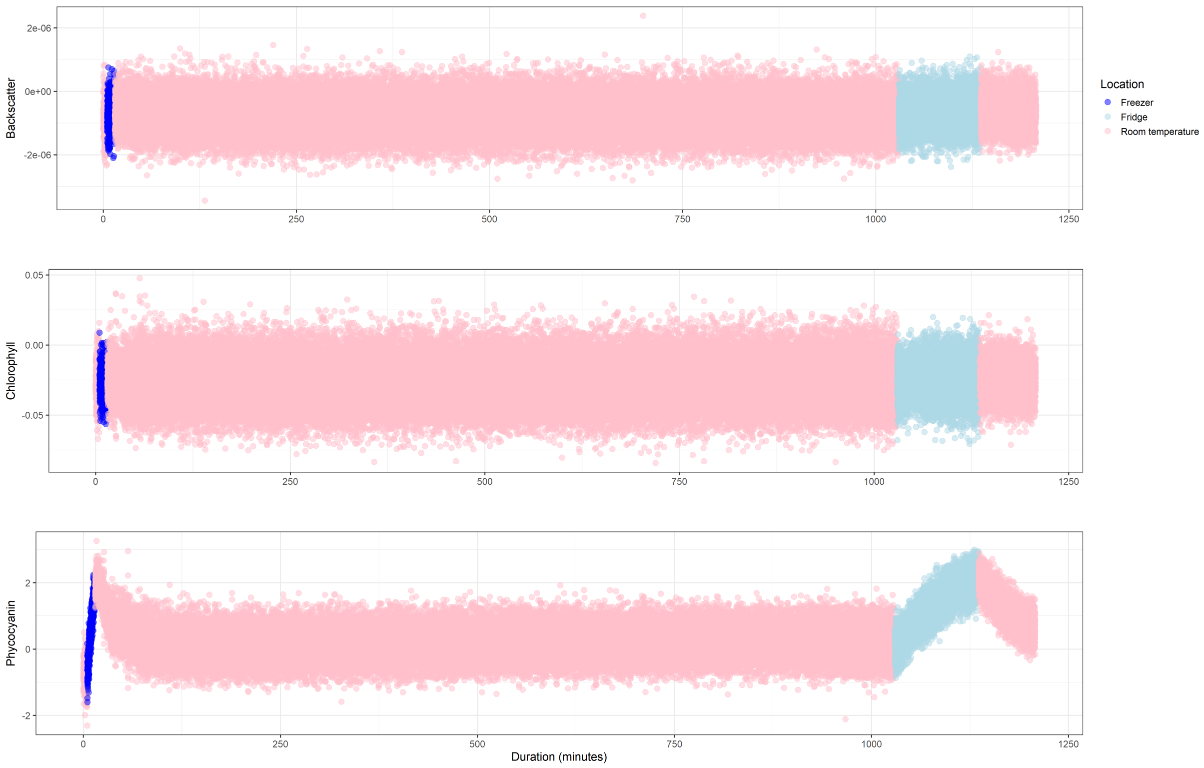

Figure 20: Serial 23833. Room temperature (20 °C), fridge (6 °C) and freezer (-18 °C). Test performed and visualized by Adele Maciute, Oceanographic Technician at Voice of the Ocean Foundation.

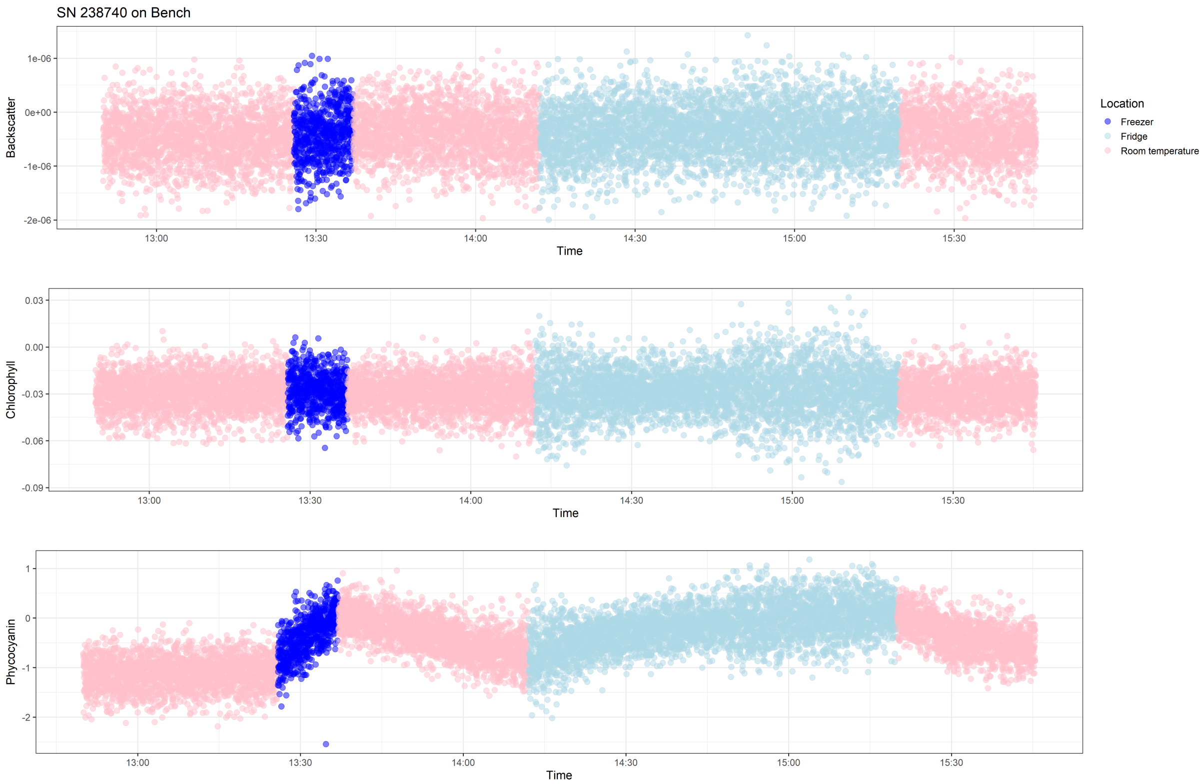

Figure 21: Serial 238740. Room temperature (20 °C), fridge (6 °C) and freezer (-18 °C). Test performed and visualized by Adele Maciute, Oceanographic Technician at Voice of the Ocean Foundation.

4 Appendix

Figure 22: Map of the Baltic Sea showing the 17 sub-basins (Map taken from http://stateofthebalticsea.helcom.fi/in-brief/our-baltic-sea/)